Bands:
Perturbation, Tearing and Breaking of Coupling Structures, and the Entropy of Changing Probabilities

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Abstract

Probabilities in practice are not fixed. They are revised as information arrives, and revised again when the information turns out to be unreliable. We model the state of knowledge as a *band*, a closed convex set of distributions, typically given by elementwise limits $(\mathrm{lo}, \mathrm{hi})$, and ask how far the algebraic and geometric structures of couplings tolerate perturbations of a band before they change. There are three regimes. Inside a chamber of margins the structures *bend*: the gluing algebra keeps its type, the coupling polytope keeps its combinatorial type, and table counts stay polynomial. On a wall they *tear*: the polytope degenerates, its vertex supports change, and for two-row tables the counts jump by explicit terms. At the boundary of the simplex they *break*: an outcome loses its support and the algebra drops dimension. The total-variation distance to tearing is $m_T = \min|f(I) - g(J)|$ and to breaking is $m_B = \min_y f_y$, both exact. Entropy is continuous in all three regimes, and the regimes are told apart by its derivatives: the product coupling's entropy is smooth everywhere inside the simplex, a vertex coupling whose cell vanishes on a wall has a $d\log(1/d)$ cusp there, and at the boundary the gradient diverges. On bands, entropy becomes an interval whose upper end is a water-filling and whose lower end sits at a vertex. Information acts by a monoid of updates on bands, not a group: conditioning commutes and is idempotent, Jeffrey updates do not commute, discounting widens bands and noisy kernels contract them. The ratio of the margin to the speed of an update is a tear time, which lower-bounds the number of updates before a structure tears. Assembling the fibres of compatible laws into families over a band gives one polytope with a computable entropy interval; limits become compatibility (cones are compatible laws) and colimits assembly of fibres; and the parameter space splits into structural shards. Along explicit paths each structure is seen to bend, tear or break. A break has second-order effects: global compatibility can be lost only through a break of the maximum-entropy filler, with an early warning that is sharp in an example; a zero at level one of the tower removes a model for ever, while reliability at level three bounds how fast the calibrated view can change and prevents the break. Two pilots on public data, NHANES subgroups and COVID-19 ensemble forecasts with their revision history, show much of the predicted behaviour: bands shrink up an information ladder, narrow structural bands must be combined with sampling bands to predict new data, decisive bands cross walls by straddling them, small margins predict tearing, and forecasts built on data that was later heavily revised lose almost half their coverage. The speed term of the tear time does

*Code, data and checks: https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/bands. `check_bands.py` (17 checks) and `check_dynamics.py` (18 checks) verify numerically the statements proved here, the synthetic validation and every quoted number; results cited from the other papers are checked there. The pilots are reproduced by the scripts in the folder `pilot`.

not yet improve on the margin. A protocol for clinical data separates the statements that are theorems from those that can fail, and it is validated on synthetic data with planted effects.

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1 Introduction

The papers of this programme treat a distribution as given: a margin f, a coupling of f with g, a posterior at some level of a tower. In practice the distribution is an estimate. It moves when new data arrive, and it moves again when old data are corrected. A forecast of a week's deaths is made four weeks ahead and revised three times before the week ends, and the counts it is scored against are themselves revised for weeks afterwards. A subgroup risk is known only within limits, and those limits narrow as more cross-tabulations are published.

This paper asks three questions. How large a change of the distributions can each structure absorb before it changes? What does entropy, as a function on sets of distributions, do as they move? And what is the right notion of the rate at which information changes them? The setting is a *band*: a closed convex set of distributions, in the simplest case all p with $\text{lo} \le p \le \text{hi}$ (section 2). A band is what the second level of the posterior tower [7] looks like from the first, and reliability is a third level: discounting a law with reliability α gives exactly the band $[\alpha p, \alpha p + 1 - \alpha]$ (proposition 2.3).

Main results.

(A) *Three regimes, with exact distances* (theorems 3.2 and 3.3). Structures bend inside a chamber, tear on a wall and break at the boundary of the simplex; chambers are taken inside the open simplices. The total-variation distance from a pair of margins to the nearest wall is $m_T = \min_{I,J} |f(I) - g(J)|$, and to the boundary is $m_B = \min_y f_y$.

(B) *Tolerance of a band* (proposition 3.4). For elementwise bands the extremes of every event probability have closed forms, so whether a band can tear is decided by finitely many sign checks.

(C) *Entropy across the regimes* (proposition 4.5). Entropy is continuous throughout. The product coupling's entropy is smooth everywhere inside the simplex; the entropy of a vertex coupling has a $d \log(1/d)$ cusp where one of its cells vanishes on a wall; at the boundary the gradient diverges. So tearing is visible to extremal structures but not to the product coupling, and breaking to both.

(D) *Entropy of a band* (propositions 4.2 to 4.4 and 4.7). The upper entropy is a water-filling, the lower entropy sits at a vertex, both move by at most $t \log(n-1) + h(t)$ when the band moves by $t \le 1 - 1/n$ in total variation, and the entropy of a mixture splits into expected entropy plus the information about which model holds. The entropy gap between the conditionally independent gluing and a joint with the same margins is the conditional mutual information.

(E) *The update monoid* (propositions 5.1 to 5.3). Updates of bands form a monoid. Conditioning on events commutes and is idempotent; Jeffrey updates do not commute, but they preserve odds ratios, so a loop of them leaves a residue that iterated fitting removes; discounting widens a band; a kernel contracts it by its Dobrushin coefficient; and conditioning can dilate a band from a point to $[0, 1]$.

(F) *Tear time* (proposition 5.8). If each update moves the margins by at most v in total variation, no structure tears before $\lceil m_T / v \rceil$ updates.

(G) *Families, limits, shards* (propositions 6.1 and 6.4 and definition 6.3). The family of fibres over a polyhedral band is a polytope whose upper entropy is the sum of the margins' upper entropies; cones are compatible laws; breaking is created only by conditioning, Jeffrey updates with a zero target and kernels with a zero column, and removed by kernels with positive entries.

(H) *Composite walls* (proposition 6.5). After a deterministic kernel the composite wall is decided by the chamber; after any other kernel it is not, by 0.59 bits on average for 2×3 margins and 5.33 of 7.94 bits for 3×3, so joint transitions need a non-deterministic kernel.

(I) *Whiskering as an update* (proposition 6.6). The whiskering defect over a band of 2×2 couplings is $t(\varepsilon)(1-2s)(1-2r)$ times a fixed circulation: zero at the unit and for the identity and swap kernels, largest at the swap coupling and for the constant kernel.

(J) *Stability along paths* (examples 7.1 and 7.2). Structure by structure, what bends, tears and breaks along a 2×2 path through two walls and along the three-bit path.

(K) *Second-order breaking* (theorem 8.1 and proposition 8.2). Compatibility is lost only through a break of the filler, and the filler's break margin divided by $\|M^+\|$ is a tolerance, attained on part of the three-bit path.

(L) *The monad of bands* (proposition 5.6). Bands are the convex powerset monad; the singleton map is a monad morphism from D whose Kleisli functor is faithful and carries every update, while any 2-functor from couplings into bands ordered by inclusion is constant on parallel kernels.

(M) *The tower under evidence* (example 9.1 and proposition 9.2). A level-one zero removes a model for ever; linear discounting with reliability α bounds the odds change per observation by $n/(1-\alpha)$ and prevents the break.

(N) *Evidence* (sections 10 and 11). Two pilots on public data, and a test protocol for clinical data in which theorems and falsifiable claims are kept apart, validated on synthetic data.

What is not claimed.

- The dynamics of information is not a Galois action. Updates do not invert, so they form a monoid; its invertible elements are relabellings, which supply symmetries and invariants but not the motion (remark 5.5).
- A loop of updates on margins does not exhibit a topological holonomy. It leaves a residue, but the updates preserve the interaction parameters, and further fitting to the original margins removes the residue (proposition 5.3). With discounting the interaction residue depends on where the discount is applied (example 5.4), but it is dissipative, not topological.
- In the forecast pilot an analogue of the tear time predicted tearing, but no better than the margin alone (section 10.2). Its speed term does not yet earn its place.
- Limits are not claimed to exist as universal cones: the category of finite probability spaces has no products, and we use cones, compatible laws, with entropy choosing a canonical one. Our shards are unrelated to the shards of hyperplane arrangements.
- In-sample containment of the true value in the NHANES bands is a sanity check, not evidence; the out-of-sample version is in section 10.1.
- In data drawn from one classical population, a common joint distribution always exists. Incompatibility of observed margins therefore measures selection between contexts, not contextuality of the kind in [4] (section 11).

Relation to the other papers. The walls and chambers are those of [4, Def. 6.3, Prop. 6.6], and the chamber-identification bound is [4, Prop. 12.2]. The gluing algebra and the table counts across walls are [1, Thm. 2.2, Prop. 7.3]. The band as a level-two shadow and the Dobrushin contraction come from [7, Prop. 9.14, Prop. 11.2]. The precision ladder of [2, Thm. 3.5] reappears as the information ladder of the NHANES pilot. The fixed-fibre geometry, constraint branes and torus branes, is in the companion note [3]; this paper supplies its dynamics. The parity horns and the signed-measure gluing are from [4, Thm. 4.3] and [6, Prop. 35.1]. Each of these papers can later gain a short subsection citing the part of this one it needs.

2 Bands

Let Y be a finite set with n elements and $\mathsf{D}(Y)$ the simplex of distributions on it.

Definition 2.1. A *band* on Y is a nonempty closed convex subset $B \subseteq \mathsf{D}(Y)$. An *elementwise band* is $B(\mathrm{lo}, \mathrm{hi}) = \{p \in \mathsf{D}(Y) : \mathrm{lo} \le p \le \mathrm{hi}\}$ for vectors $0 \le \mathrm{lo} \le \mathrm{hi} \le 1$; it is nonempty if and only if $\sum \mathrm{lo} \le 1 \le \sum \mathrm{hi}$. Its *tightening* is

$$\mathrm{lo}'_y = \max\Big(\mathrm{lo}_y,\ 1 - \sum_{z \ne y} \mathrm{hi}_z\Big), \qquad \mathrm{hi}'_y = \min\Big(\mathrm{hi}_y,\ 1 - \sum_{z \ne y} \mathrm{lo}_z\Big).$$

Elementwise bands are the probability intervals of de Campos, Huete and Moral [14], and general bands are credal sets in the sense of Walley [29].

Proposition 2.2 (Tightening and event probabilities)**.** *Let $B = B(\mathrm{lo}, \mathrm{hi})$ be nonempty. Then $B(\mathrm{lo}', \mathrm{hi}') = B$, every tightened bound is attained by some $p \in B$, and for every event $I \subseteq Y$,*

$$\min_{p \in B} p(I) = \max\Big(\sum_{y \in I} \mathrm{lo}_y,\ 1 - \sum_{y \notin I} \mathrm{hi}_y\Big), \qquad \max_{p \in B} p(I) = \min\Big(\sum_{y \in I} \mathrm{hi}_y,\ 1 - \sum_{y \notin I} \mathrm{lo}_y\Big).$$

Proof. For $p \in B$, $p(I) = 1 - p(I^c) \ge 1 - \sum_{I^c} \mathrm{hi}$ and $p(I) \ge \sum_I \mathrm{lo}$, so the stated minimum is a lower bound. It is attained by setting $p = \mathrm{hi}$ on I^c and $p = \mathrm{lo}$ on I and then distributing the remaining mass $1 - \sum_I \mathrm{lo} - \sum_{I^c} \mathrm{hi}$, which has the sign that makes this possible, on I if it is positive and on I^c if it is negative, within the bounds; this is a greedy fill, possible because $\sum \mathrm{lo} \le 1 \le \sum \mathrm{hi}$. The maximum is symmetric. Singletons $I = \{y\}$ give the tightened bounds, which are therefore attained, and they cut out the same set. □

Proposition 2.3 (Reliability gives a band)**.** *Let $p \in \mathsf{D}(Y)$ and $\alpha \in [0, 1]$. The set $\{\alpha p + (1 - \alpha)q : q \in \mathsf{D}(Y)\}$ of laws obtained by trusting p with reliability α and allowing anything otherwise, the credal set of Shafer's discounting [27], is exactly the elementwise band $B(\alpha p,\ \alpha p + (1 - \alpha)\mathbf{1})$.*

Proof. Each $\alpha p + (1 - \alpha)q$ lies between αp and $\alpha p + (1 - \alpha)$ coordinatewise. Conversely, if r lies in the band and sums to 1, then $q = (r - \alpha p)/(1 - \alpha)$ is nonnegative and sums to 1 (for $\alpha < 1$; for $\alpha = 1$ both sets are $\{p\}$). □

Bands from the tower. A finitely supported law $\Pi = \sum_k w_k \delta_{\rho_k}$ on $\mathsf{D}(Y)$, a level-two element of the tower [7], generates the band $\mathrm{conv}\{\rho_k\}$, which contains the barycenter $\sum_k w_k \rho_k$. The band forgets the weights w_k, and that is the precise sense in which it is the level-one shadow of level two. A level-three law, a law on such Π, models doubt about the reliability of the level-two assessment; proposition 2.3 is the simplest instance, with Π supported on $\{p\}$ and the discounting weight standing for the doubt.

Banded margins. When both margins are only known to lie in bands, the cells of their couplings are bounded by the Fréchet bounds taken at the extremes of the bands.

Proposition 2.4 (Fréchet bounds over bands)**.** *Let $F = B(\mathrm{lo}^f, \mathrm{hi}^f)$ and $G = B(\mathrm{lo}^g, \mathrm{hi}^g)$ be tightened. Over all $f \in F$, $g \in G$ and $\alpha \in \mathrm{Cpl}(f, g)$,*

$$\max\big(0,\ \mathrm{lo}^f_i + \mathrm{lo}^g_j - 1\big) \ \le\ \alpha_{ij} \ \le\ \min\big(\mathrm{hi}^f_i,\ \mathrm{hi}^g_j\big),$$

and both bounds are attained.

Proof. For fixed margins the Fréchet bounds $\max(0, f_i + g_j - 1) \le \alpha_{ij} \le \min(f_i, g_j)$ are sharp. They are monotone in f_i and g_j, and the two bands are independent, so the extremes are attained at f_i, g_j both minimal or both maximal, which is possible by proposition 2.2. □

3 Three regimes

Let f and g be distributions on sets with n and m elements. A *wall* is a hyperplane $f(I) = g(J)$ with I, J nonempty and proper, and the *chambers* are the components of the complement of the walls in the product of the open simplices [4, Def. 6.3]. Off the open simplices the polytope can degenerate without any wall, which is the breaking regime.

Definition 3.1. The *tear margin* of (f, g) is $m_T(f, g) = \min_{I,J} |f(I) - g(J)|$, over nonempty proper I, J. The *break margin* of f is $m_B(f) = \min_y f_y$, and of the pair it is $\min(m_B(f), m_B(g))$. Distances between pairs are measured by $\mathrm{TV}(f, f') + \mathrm{TV}(g, g')$, with $\mathrm{TV}(p, q) = \frac{1}{2}\sum |p - q|$.

Theorem 3.2 (Exact distances)**.**

(i) The distance from (f, g) to the union of the walls is $m_T(f, g)$.

(ii) The distance from f to the face $\{p_y = 0\}$ of the simplex is f_y, so the distance to its boundary is $m_B(f)$.

Proof. (i) For any event, $|f(I) - f'(I)| \le \mathrm{TV}(f, f')$, so $f(I) - g(J)$ changes by at most $\mathrm{TV}(f, f') + \mathrm{TV}(g, g')$ and no pair closer than m_T lies on a wall. Conversely, if $f(I) - g(J) = d > 0$, moving mass d from I to I^c in f costs exactly d in total variation and lands on the wall; this is possible because $f(I) \ge d$ and $I^c \ne \emptyset$. (ii) Moving the mass f_y elsewhere costs f_y, and any p with $p_y = 0$ has $\mathrm{TV}(f, p) \ge f_y$. □

By [4, Prop. 12.2], $N \ge 2\log(2W/\eta)/m_T^2$ samples identify the chamber with probability at least $1 - \eta$, where W is the number of walls. So the tear margin is at once a geometric distance and a sample size.

Theorem 3.3 (Bending, tearing, breaking)**.** *(i)* Bending. *On the open simplex the gluing algebra A_f is isomorphic to $\mathbb{R} \times M_{n-1}(\mathbb{R})$ by an isomorphism that depends continuously on f [1, Thm. 2.2]. On a chamber the coupling polytope* $\mathrm{Cpl}(f, g)$ *is simple and of constant combinatorial type [4, Thm. 6.4, Prop. 6.6]; its vertices have constant supports, the spanning trees of the complete bipartite graph that are feasible, each vertex is an affine function of (f, g) with exactly $n + m - 1$ positive cells, and for two-row tables the number of integer tables is a single polynomial [1, Prop. 7.3].*

(ii) Tearing. *On a wall the coupling polytope is degenerate: some vertex has fewer than $n + m - 1$ positive cells [15]. Crossing the wall changes the set of feasible spanning trees, and with it the labelled structure of the vertices; the number of vertices may or may not change (for 2×2 the polytope is a segment on both sides). For $2 \times m$ tables the change in the count of integer tables is one binomial term, and a corner cut of lattice depth d removes $\binom{d+1}{2}$ tables [1, Prop. 7.3]; in the 2×3 case of [4, Prop. 10.6] the area changes by $\frac{1}{2}s^2$.*

(iii) Breaking. *When $f_y \to 0$, the gluing product $a \cdot b = a\,\mathrm{diag}(f)^{-1} b$ involves $\mathrm{diag}(f)^{-1}$ of norm $1/m_B$, and the Fisher information of the multinomial family, $\mathrm{diag}(f)^{-1}$ on the tangent space, diverges. On the face $f_y = 0$ the algebra is replaced by $A_{f|_S}$ for the support S, of dimension $(|S| - 1)^2 + 1$, strictly smaller.*

Proof. (i) The isomorphism is the congruence $a \mapsto \mathrm{diag}(f)^{-1/2} a\,\mathrm{diag}(f)^{-1/2}$. For nondegenerate margins the transportation polytope is simple, and its vertices are the basic feasible solutions, whose supports are spanning trees of the complete bipartite graph. Removing an edge of a tree splits its nodes into a part containing rows I and columns J and the rest, and the flow on that

edge is $\pm(f(I) - g(J))$, the sign depending on whether the edge leaves the part from a row or from a column. So which trees are feasible is decided by the signs of the wall functions, which are constant on a chamber, and each basic solution solves a fixed linear system in (f, g). (ii) The same computation shows that on a wall some edge of some feasible tree carries flow 0, so the corresponding vertex is degenerate; that degeneracy occurs exactly on the walls is classical [15]. The jumps are the cited results. (iii) Direct; the dimension formula is [1, Thm. 2.2] applied to the support. □

Proposition 3.4 (Tolerance of a band)**.** *Let F and G be elementwise bands for f and g. The pair of bands meets no wall if and only if, for every wall (I, J), either $\min_F f(I) > \max_G g(J)$ or $\max_F f(I) < \min_G g(J)$, with the extremes given by proposition 2.2. It meets no face of the simplex if and only if all tightened lower bounds are positive.*

Proof. The bands are independent, so the range of $f(I) - g(J)$ over $F \times G$ is the interval between $\min f(I) - \max g(J)$ and $\max f(I) - \min g(J)$, and it avoids 0 exactly under the stated condition. The face statement is proposition 2.2 for singletons. □

So the tolerance of a band is decided by $2(2^{n-1} - 1)(2^m - 2)$ closed-form numbers, one minimum and one maximum per wall, and the margin by which it is safe is the smallest gap among them.

Remark 3.5 (Walls are GIT walls). The coupling polytope is a fibre of the margin map $M : \mathbb{R}^{n \times m}_{\geq 0} \to \mathbb{R}^{n+m}$, whose columns are the vectors $e_i + e_{n+j}$. The torus of row and column scalings, modulo the circle $(c, \dots, c, c^{-1}, \dots, c^{-1})$ that acts trivially, acts on $\mathbb{C}^{n \times m}$ with these weights, and the toric variety $X_{f,g}$ of [4, Thm. 6.4] is its symplectic reduction at level (f, g), equivalently the GIT quotient linearized by the character (f, g). Moving (f, g) is therefore variation of GIT quotients [32, 31], whose walls are the hyperplanes spanned by subsets of the weights of codimension one; these are exactly the walls $f(I) = g(J)$ together with the faces $f_i = 0$ and $g_j = 0$ [4, Prop. 10.5], which we recheck by direct enumeration for 2×2, 2×3 and 3×3 (section 13). So tearing is a GIT wall-crossing: on the wall the transportation polytope is degenerate (theorem 3.3(ii)) and the quotient has strictly semistable points; the quotient at the wall is the common contraction of the quotients on the two sides, smooth at a blow-up wall, as for 2×3, and singular at a flip wall, as in many 3×3 crossings [4, §10.2]; for 2×2 all three are $\mathbb{CP}^1$. Breaking is reaching the boundary of the cone of characters, where the quotient loses dimension. Neither is an obstruction class in the cohomological sense. The first-order deformations of a coupling with fixed margins are the zero-margin matrices, a linear space of dimension $(n-1)(m-1)$, so nothing is obstructed, and the gluing algebra is rigid, $\mathrm{HH}^{>0}(A_f) = 0$ [1]. The regimes are combinatorial: chambers, walls, and the boundary of the cone.

4 Entropy on bands and across the regimes

Write $H(p) = -\sum p_y \log p_y$ and $h(t) = H(t, 1-t)$.

Definition 4.1. The *upper* and *lower entropy* of a band B are $\overline{H}(B) = \max_B H$ and $\underline{H}(B) = \min_B H$; their difference is the entropy that is due to imprecision rather than to randomness.

Proposition 4.2 (Upper entropy is a water-filling)**.** *For an elementwise band, $\overline{H}$ is attained at the unique point $p_y = \mathrm{clip}(c, \mathrm{lo}_y, \mathrm{hi}_y)$, where c is the level at which these values sum to 1.*

Proof. H is strictly concave, so the maximizer is unique and characterized by the Karush–Kuhn–Tucker conditions $-\log p_y - 1 - \lambda + \mu_y - \nu_y = 0$ with $\mu_y \geq 0$ active only at lo_y and $\nu_y \geq 0$ active

only at hi_y. So $p_y = e^{-1-\lambda}$ wherever no bound is active, $p_y = \mathrm{lo}_y$ only if $e^{-1-\lambda} \le \mathrm{lo}_y$, and $p_y = \mathrm{hi}_y$ only if $e^{-1-\lambda} \ge \mathrm{hi}_y$; with $c = e^{-1-\lambda}$ this is the clip. Its sum is continuous and nondecreasing in c, so c exists. Algorithms of this kind for credal sets are given by Abellán and Moral [8]. □

Proposition 4.3 (Lower entropy sits at a vertex)**.** *$\underline{H}$ of a polytope band is attained at a vertex. The vertices of an elementwise band are the points with every coordinate but at most one at a bound, so there are at most $n2^{n-1}$ of them.*

Proof. A concave function attains its minimum over a polytope at a vertex. A point of $B(\mathrm{lo}, \mathrm{hi})$ with two coordinates strictly inside their bounds lies on a segment in B obtained by trading mass between them, so it is not a vertex. □

Proposition 4.4 (Entropy moves slowly)**.** *If $\mathrm{TV}(p, q) = t \le 1 - 1/n$, then $|H(p) - H(q)| \le t \log(n-1) + h(t)$, with equality for $p = \delta_y$ and $q = (1-t, \frac{t}{n-1}, \dots)$ [30, 11]. Consequently, if two bands are within Hausdorff distance t in total variation with $t \le 1 - 1/n$, their upper entropies and their lower entropies each differ by at most $t \log(n-1) + h(t)$.*

Proof. The first statement is the sharp continuity bound. The bound is increasing in t on $[0, 1-1/n]$, since its derivative $\log(n-1) + \log \frac{1-t}{t}$ is nonnegative there. For every $p \in B$ there is $p' \in B'$ within t, so $\overline{H}(B') \ge H(p) - t \log(n-1) - h(t)$; take the maximum over p, and argue symmetrically and likewise for minima. □

So an update that moves a band by t in total variation changes its entropy interval by at most $t \log(n-1) + h(t)$: the rate of change of entropy is controlled by the rate of change of the band.

Proposition 4.5 (Entropy across the regimes)**.** *(i) The entropy $H(f \otimes g) = H(f) + H(g)$ of the product coupling, the maximum-entropy coupling, is real-analytic on the product of the open simplices. It does not see walls.*

(ii) Let $v(f, g)$ be a vertex of $\mathrm{Cpl}(f, g)$ on a chamber, and suppose that, as (f, g) approaches a point of a wall along a segment, one of its cells tends to 0 while the others stay positive. With t the distance along the segment, $H(v(t)) = H(v(0)) + \kappa t \log(1/t) + O(t)$ for some $\kappa > 0$, so its derivative in t diverges like $\kappa \log(1/t)$: a cusp.

(iii) As $f_y \to 0$, $\partial H(f)/\partial f_y = -\log f_y - 1$ diverges, and so does the corresponding derivative of the product-coupling entropy $H(f) + H(g)$.

So entropy is continuous in all three regimes. For a fixed vertex, bending is smooth, tearing is a logarithmic cusp, and breaking is a divergent gradient; the product coupling sees only the last.

Proof. (i) is immediate. (ii) On the chamber v is affine in (f, g) by theorem 3.3(i), so the vanishing cell is κt with $\kappa > 0$, and $-\kappa t \log(\kappa t) = -\kappa t \log t + O(t)$; the other cells stay positive, so their terms are smooth. (iii) Direct. □

Example 4.6. For 2×2 couplings with $g = (0.3, 0.7)$, the comonotone vertex has $\alpha_{11} = \min(f_1, 0.3)$. Its entropy has one-sided slope 7.55, 12.16 and 16.76 at distances 10^{-3}, 10^{-5} and 10^{-7} past the wall $f_1 = 0.3$, growing by $\log 100$ per factor 100, while the entropy of the product coupling has equal one-sided slopes there (check B7).

The lower entropy of the coupling polytope, the minimum over its vertices, is a minimum of finitely many analytic functions on each chamber. It therefore has ordinary kinks inside chambers where the minimizing vertex changes: for $g = (0.3, 0.7)$ the comonotone and countermonotone

entropies cross at $f_1 = 0.5$, inside a chamber. It has a logarithmic cusp at a wall only when a minimizing vertex degenerates there, and it can be smooth across a wall otherwise. What distinguishes tearing is therefore the $\log(1/t)$ blow-up of a slope, not non-smoothness as such. The upper end of the entropy interval of the couplings, the product coupling, is blind to walls altogether.

Proposition 4.7 (Entropy of a mixture)**.** *For $\Pi = \sum_k w_k \delta_{\rho_k}$ with barycenter $\bar{\rho} = \sum_k w_k \rho_k$,*

$$H(\bar{\rho}) = \sum_k w_k H(\rho_k) + I(K; X),$$

where (K, X) has law $w_k \rho_k(x)$. The first term is the expected entropy of the outcome under each model, the aleatoric part, and the second is the information the outcome carries about the model, the epistemic part.

Proof. $H(X) = H(X \mid K) + I(K; X)$. □

Proposition 4.8 (The gluing gap)**.** *Let P be a law of (X, Y, Z) and $F = P_{XY} P_{YZ} / P_Y$ the conditionally independent gluing of its margins, the canonical filler of [4, Thm. 4.2]. Then F has the same (X, Y) and (Y, Z) margins as P, and*

$$H(F) - H(P) = I_P(X; Z \mid Y) \ \geq 0.$$

Proof. $H(F) = H(X, Y) + H(Y, Z) - H(Y)$, and $H(P) = H(X, Y) + H(Y, Z) - H(Y) - I_P(X; Z \mid Y)$. □

So the filler maximizes entropy in its fibre for every P. What can be tested on data is not that, but the size of the gap.

5 The update monoid

An *update* is a partial map from bands to bands: it sends a band B to the closed convex hull of the images of its members, and it is defined when it is defined on every member. We consider four kinds: conditioning on an event E, $p \mapsto p(\cdot \mid E)$, defined when $p(E) > 0$ throughout B; the Jeffrey update on a partition $\{E_i\}$ to target weights q, $p \mapsto \sum_i q_i p(\cdot \mid E_i)$ [16], defined when every $p(E_i) > 0$; discounting by reliability α, which sends p to the band of proposition 2.3; and pushforward by a kernel. The hull is needed because the image of a convex band under a Jeffrey update need not be convex. Composition makes the updates a monoid of partial maps.

Proposition 5.1 (Properties of the monoid)**.** *(i) Conditioning on events commutes, $p(\cdot \mid E)(\cdot \mid E') = p(\cdot \mid E \cap E')$, and is idempotent.*

(ii) Jeffrey updates on two partitions do not commute in general. They commute at a law under which the two partitions are independent.

(iii) Discounting sends an elementwise band of width w_y at y to one of width $\alpha w_y + 1 - \alpha$. A kernel K contracts the total-variation diameter of every band by at least its Dobrushin coefficient $\eta(K) = \max_{y,y'} \mathrm{TV}(K_y, K_{y'})$.

(iv) The invertible kernels are the permutations of Y. In the monoid $\mathrm{End}(f)$ of kernels that preserve f, the invertible elements are the permutations σ with $f \circ \sigma = f$.

Proof. (i) is direct. (ii) For the 2×2 law with rows $(0.30, 0.10)$ and $(0.15, 0.45)$, updating rows to $(0.6, 0.4)$ and columns to $(0.35, 0.65)$ in the two orders gives laws at total variation 0.120 (check B9). If the partitions are independent, the law of the two partitions is preserved by both updates, and for $y \in E_i \cap F_j$, with all $p(E_i), p(F_j) > 0$, either order gives $q_i r_j \, p(y)/(p(E_i)p(F_j))$, which is symmetric in the two updates. (iii) The first statement follows from proposition 2.3, applied to the extreme points. For the second, $\mathrm{TV}(K^{\mathsf{T}}p, K^{\mathsf{T}}q) \le \eta(K)\mathrm{TV}(p, q)$ [17], as in [7, Prop. 11.2]. (iv) A stochastic matrix with a stochastic inverse is a permutation matrix, and it preserves f exactly when $f \circ \sigma = f$; $\mathrm{End}(f)$ is the monoid of [1, Prop. 3.1]. □

Proposition 5.2 (Dilation)**.** *Let B be the band of laws of $(X, Y) \in \{0,1\}^2$ with $P(X = 1) = P(Y = 1) = \frac{1}{2}$. The band for $P(X = 1)$ is the point $\frac{1}{2}$, but after conditioning on $Y = y$ it is $[0, 1]$, for both values of y.*

Proof. $P(X = 1 \mid Y = 1) = 2p_{11}$ with $p_{11} \in [0, \frac{1}{2}]$ free, and likewise for $Y = 0$. □

This is the phenomenon of Seidenfeld and Wasserman [26]: learning something can make every possible conclusion less certain. It is how unreliable information enters the formalism.

Proposition 5.3 (Loops of Jeffrey updates)**.** *On positive two-way tables, Jeffrey updates on the row partition and on the column partition preserve every odds ratio $p_{ij}p_{kl}/(p_{il}p_{kj})$. A loop that moves the row margin to q_1, the column margin to r_1, and then back to the original q_0 and r_0 in general does not return to the starting law. Iterating the last two updates converges back to it.*

Proof. A row update multiplies each row by a constant and a column update each column, and both leave cross ratios unchanged. For the law in the proof of proposition 5.1, the loop ends at total variation 0.073 from the start, with the odds ratio 9 kept. Alternating the two updates is iterative proportional fitting, which converges to the unique law with the target margins and the given odds ratios, the information projection of Csiszár [13]; the starting law is that law. □

So the residue of a loop persists if nothing more is done, but it lies entirely in the margins: the updates leave the interaction alone, and refitting the original margins removes it. A loop that changes an invariant needs updates that act on interactions. Linear discounting towards the uniform law is one, and with it the residue in the interaction depends on where the discount is applied.

Example 5.4 (A path-dependent residue)**.** Start from the law of proposition 5.3, with odds ratio 9, and run the loop that refits the margins to $(0.8, 0.2)$ and r_0, then to $(0.8, 0.2)$ and $(0.2, 0.8)$, then to q_0 and $(0.2, 0.8)$, then back to q_0 and r_0, applying one linear discount with reliability 0.9 after one of the four corners. The final odds ratio is 4.63, 2.22, 5.93 or 6.89 according to the corner; with discounts after the first and third corners it is 3.64, after the second and fourth 2.05; without discounting the loop returns exactly to 9. The same updates in the same number, placed at different points of the loop, leave different interactions, because a discount shrinks the odds ratio by an amount that depends on the margins at that moment, and refitting keeps odds ratios. So the residue depends on where along the path the discount is applied, not on the loop as such: going straight to the second corner, discounting and coming straight back gives the same 2.22. It is a dissipative residue of discounting not commuting with refitting, not a topological holonomy, and refitting the margins does not remove it.

Remark 5.5 (Not a Galois action). The motion of information is not a Galois action, because updates cannot be undone: conditioning forgets and discounting widens. A Galois-type structure does appear in two places. The units of the update monoid are the relabellings of Y, and those

that preserve f, the units of $\mathrm{End}(f)$, act on the structures over f and preserve every quantity in this paper; their subgroups correspond to the partitions they preserve by a Galois connection. And the fibre-level counterpart of a Galois action, monodromy, is the loop question of proposition 5.3.

5.1 The monad of bands, and why it stops at the 1-cells

Bands carry a monad of their own. For a finite set Y let $\mathsf{C}(Y)$ be the set of nonempty finitely generated convex subsets of $\mathsf{D}(Y)$, the polytope bands, elementwise bands among them. With unit $y \mapsto \{\delta_y\}$, and Kleisli extension of a band-valued map $\varphi\colon Y \to \mathsf{C}(Y')$ given on a band B by

$$\varphi^{\#}(B) = \Big\{ \textstyle\sum_y p_y q_y \;:\; p \in B,\; q_y \in \varphi(y) \Big\},$$

C is the convex powerset monad, which combines probability with nondeterminism; there is no distributive law of D over the powerset monad, and C arises instead from a weak distributive law [28, 20, 18].

Proposition 5.6 (Bands as a monad)**.** *(i) $p \mapsto \{p\}$ is a morphism of monads $\mathsf{D} \to \mathsf{C}$, so it induces a faithful functor J from the Kleisli category of D, the kernels, to that of C, the band-valued kernels, which is the identity on objects.*

(ii) Kernel updates, Jeffrey updates and, where defined, conditioning act on bands as $J(K)^{\#}$; linear discounting is J of a kernel, and Shafer discounting $p \mapsto B(\alpha p, \alpha p + 1 - \alpha)$ is a band-valued map not in the image of J. So the update monoid of section 5 is a monoid of Kleisli maps of C.

(iii) The Kleisli category of C is locally posetal: $\varphi \le \varphi'$ when $\varphi(y) \subseteq \varphi'(y)$ for all y, and composition is monotone.

(iv) Any 2-functor, strict or lax, from a 2-category whose 1-cells are kernels and whose 2-cells are couplings to a locally posetal 2-category identifies all parallel kernels. For any two parallel kernels f, g there are couplings $f \Rightarrow g$ and $g \Rightarrow f$, the product coupling and its transpose, so the images satisfy $F(f) \le F(g) \le F(f)$.

Proof. (i) The singleton map commutes with the units, and with the multiplications because $\varphi^{\#}(\{p\})$ for singleton-valued φ is the singleton of the Kleisli extension in D. Faithfulness: distinct kernels have distinct singleton values. (ii) Pointwise application to the members of a band, followed by the convex hull, is $J(K)^{\#}$; the extension formula takes convex combinations, so the hull is automatic for polytope bands. Shafer discounting has non-singleton values. (iii) The extension formula is monotone in B and in each $\varphi(y)$. (iv) A 2-functor sends a 2-cell $f \Rightarrow g$ to a 2-cell $F(f) \Rightarrow F(g)$, which in a locally posetal 2-category is an inequality. □

So the bridge from the monad D to the band picture is a functor at the level of 1-cells, J, and it is as good as one could ask: it is faithful and carries every update. At the level of 2-cells, a strict or lax 2-functor from the couplings into the bands, ordered by inclusion, can only be constant, because couplings exist in both directions between any two kernels. Couplings have to be the hom-objects themselves: each $\mathrm{Cpl}(f, g)$ is a polytope, gluing is biaffine on it, and the shards of section 6 and the fibres of proposition 6.1 are bands of 2-cells in this sense. Whiskering is where this enriched structure stops being strict (proposition 6.6).

Definition 5.7. For a sequence of margin pairs in which each update moves the pair by at most v in the distance of definition 3.1, the *tear time* is $\tau = m_T/v$, and the *break time* is m_B/v.

Proposition 5.8 (Tear time)**.** *No structure of theorem 3.3 tears before $\lceil m_T/v \rceil$ updates, and none breaks before $\lceil m_B/v \rceil$ updates.*

Proof. By theorem 3.2 the walls are at distance m_T and the boundary at distance m_B, and each update covers at most v. □

The tear time is only a lower bound, because updates need not head for the nearest wall. Whether a ratio of margin to speed is a useful predictor is an empirical question, and section 10.2 gives a mixed answer for a one-dimensional analogue.

6 Families, limits and shards

The paper so far moves one law, or one pair of margins, at a time. The structures of [3] live on fibres: for marginal data m, the fibre $C_m = \{\pi \in \mathsf{D}(X) : M\pi = m\}$ of compatible joint laws, its maximum-entropy point and its toric and brane data. This section assembles the fibres into families, says what limits and colimits mean for them, and cuts the parameter space into regimes.

6.1 The family over a band

Proposition 6.1 (Families over bands)**.** *Let B be a band of marginal data.*

(i) The family $\mathcal{P}_B = \bigcup_{m \in B} C_m$ equals $\{\pi \in \mathsf{D}(X) : M\pi \in B\}$, the preimage of B; it is a polytope when B is.

(ii) For couplings whose margins lie in elementwise bands F and G, the upper entropy of $\mathcal{P}_B$ is $\overline{H}(F) + \overline{H}(G)$, attained at the product of the two water-fillings of proposition 4.2; the lower entropy is attained at a vertex.

Proof. (i) is the definition of a preimage, and the preimage of a polytope under a linear map, intersected with the simplex, is a polytope. (ii) $H(\alpha) \le H(f) + H(g)$ for any coupling of f and g, with equality at the product, and the product of the water-fillings lies in $\mathcal{P}_B$. The lower entropy is a concave minimum over a polytope. □

So the family is not merely a union of polytopes: over a polyhedral band, for example an elementwise one, it is one polytope, and over any convex band it is convex; its entropy interval is computable. In the lattice of subsets of $\mathsf{D}(X)$, a category different from the one used for limits below, $\mathcal{P}_B$ is the colimit, the union, of the fibres C_m over $m \in B$: assembly of the fibres.

6.2 Limits as compatibility

Let local laws m_S on X_S be given for the sets S of a cover, agreeing on overlaps. A *cone* over this diagram, in the category of finite probability spaces and measure-preserving maps, is a probability space with measure-preserving maps to the local spaces, commuting with the marginal maps. Every cone pushes forward to one with apex X and the coordinate projections as legs, and those are exactly the joint laws π with $\pi_S = m_S$: the fibre C_m. The category has no products, and in general there is no universal cone, so "the limit exists" has to be read as "a cone exists": compatibility, or a global section in the sense of Abramsky and Brandenburger [10]. Two refinements make it quantitative. Positive cones exist iff $C_m^\circ \neq \emptyset$, and then entropy selects a canonical cone, the maximum-entropy law, which is where the entropy brane meets the conormal brane of the constraint [3, Thm. 3.1].

When no cone exists, the contextual fraction [9] measures the failure as one minus the largest weight λ of a sub-normalized global section, $\lambda\pi_S \leq m_S$ for all S.

Whether a cone exists depends on the shape of the cover, not on the monad: the laws of D hold identically, whatever the data. By Vorob'ev's theorem [33], compatible laws on an acyclic cover always glue, so failure needs a cycle. The following proposition shows where the first failures sit in simplicial terms. Its horn statements are those of the coupling nerve in [4, Thm. 4.2, Thm. 4.3, Cor. 4.4], where the parity horns and the contextual fraction of a horn are also computed; what is added here is the contrast with boundaries in dimension 2 and an explicit 3-horn built from the three-bit model. The *marginal nerve* of a finite family of finite variables is the semi-simplicial set whose k-simplices are the laws on sets of $k+1$ of the variables, with marginals as face maps. A *k-horn* is a set of laws on all the k-element subsets of a $(k+1)$-element set S but one, agreeing on overlaps; a *k-boundary* has laws on all of them. A *filler* is a law on X_S with the given marginals.

Proposition 6.2 (Horns and boundaries)**.**

(i) Every 2-horn fills: given p_{ab} and p_{bc} agreeing on b, the law $p_{ab}(a,b)\,p_{bc}(b,c)/p_b(b)$, set to 0 where $p_b(b) = 0$, is a filler.

(ii) A 2-boundary need not fill. Three binary variables that are pairwise uniform and perfectly anticorrelated have no joint law. This is the three-bit model of [3, Ex. 3.3] at $\varepsilon = 0$, the minimal contextual configuration.

(iii) A 3-horn need not fill. Let X be uniform and independent of each pair, and Y, Z, W pairwise uniform and perfectly anticorrelated. The laws on XYZ, XYW and XZW are valid and agree on overlaps, and there is no law on $XYZW$.

So the marginal nerve is not Kan [4, Cor. 4.4]. But this is a consequence of contextuality, not its definition: the minimal contextual configuration is a boundary in dimension 2, where every horn fills. Contextuality is the failure to fill boundaries over cyclic covers; the Kan condition only records failure from dimension 3 on, and records it as yes or no, while the contextual fraction measures it.

Proof. (i) The stated law has the right marginals by direct summation; the two edges sharing a vertex form an acyclic cover. (ii) Three binary variables cannot be pairwise different. (iii) Each given face is the product of a uniform X and a perfectly anticorrelated uniform pair, so it is a law, and the overlaps XY, XZ, XW are all uniform products. A filler would have $Y \neq Z$, $Y \neq W$ and $Z \neq W$ almost surely, which is (ii). □

6.3 Shards

Definition 6.3 (Structural shards)**.** Let Θ be a space of parameters θ: marginal data m, possibly a pair of margins (f, g) or a band of them, and possibly a kernel. Its *signature* $\Sigma(\theta)$ has up to three components, each defined when the parameter has the relevant part:

- σ_{pos}: whether compatible laws exist and, if so, the support of a law in the relative interior of C_m (the smallest face of the simplex containing C_m); for a band, the set of such supports over its members;
- σ_{wall}, for pairs of margins: the vector of signs of the wall functions $f(I) - g(J)$, with 0 on a wall; for bands, each entry is above, below or straddling, as in proposition 3.4;
- σ_{comp}, for kernels: deterministic, constant but not deterministic, or neither.

A *structural shard* is a connected component of a level set of Σ. The *interaction nerve* has the shards as vertices and a simplex for every set of shards whose closures share a point.

The name is chosen for its picture, and the notion is unrelated to the shards of hyperplane arrangements [24]. The stratification is multiaxial: the three components of Σ change independently, so a parameter can sit in the interior of a chamber and still carry a stochastic kernel. Because σ_{pos} records the support, not only whether a positive law exists, the dimension of the fibre and of the gluing algebra is constant on a shard. On a shard the structures of theorem 3.3 bend; between shards they tear or break; and the kernel class decides which compositions are geometric [3, Thm. 6.1].

Proposition 6.4 (Which updates move between shards)**.** *(i) Conditioning on an event E with $p(E^c) > 0$ breaks: it sets the probabilities outside E to 0.*

(ii) A Jeffrey update keeps the support of the law if all its targets are positive, and breaks if some target is 0.

(iii) Shafer discounting, which sends p to the band $B(\alpha p, \alpha p + 1 - \alpha)$ of proposition 2.3, keeps αp as the lower envelope, so it neither breaks nor heals the lower envelope. Linear discounting towards the uniform law, $p \mapsto \alpha p + (1-\alpha)u$, heals: afterwards every probability is at least $(1-\alpha)/n$.

(iv) A kernel sends a full-support law to a full-support law iff no column of the kernel is zero, and sends every law to a full-support law if all its entries are positive.

(v) Jeffrey updates, discounting and kernels can cross walls.

Proof. (i)–(iv) are direct; in (iv), $(K^{\mathsf{T}}p)_j = \sum_y p_y K_{yj}$, and linear discounting is the kernel $\alpha I + (1-\alpha)J/n$, with all entries positive. For (v), these updates move the margins continuously in their parameters (targets, reliability, kernel entries), and the wall functions are continuous; for instance a kernel moves a margin g to $K^{\mathsf{T}}g$, and for suitable K this satisfies $f(I) = (K^{\mathsf{T}}g)(J)$. □

So among these updates, breaking is created only by conditioning, by Jeffrey updates with a zero target and by kernels with a zero column, and it is removed by kernels whose entries are positive, linear discounting towards the uniform law among them. Composition with a kernel that has no zero column can tear but never break, and in the 2×2 case it also contracts the moment segment by the Dobrushin coefficient [3, Ex. 6.2]. Where the axes interact is visible here: a stochastic composition, an event of the σ_{comp} axis, moves the margins and can land them on a wall, an event of the σ_{wall} axis.

6.4 The wall and kernel axes interact only through stochastic kernels

Compose 2×3 couplings with a kernel $K\colon \{0,1,2\} \to \mathsf{D}(\{0,1\})$. The parameters are (f, g, K), the original margins have the wall arrangement of $f(I) = g(J)$, and the composite 2×2 coupling, with margins f and $h = K^{\mathsf{T}}g$, has the single wall $f_1 = h_1$; here f_1 and h_1 are the probabilities of the first outcome, which carries the label 0. Call the sign of $f_1 - h_1$ the *composite side*.

Proposition 6.5 (Composite walls)**.** *If K is deterministic, a function ϕ, the composite side is determined by the chamber of (f, g) and ϕ: it is the sign of the original wall function $f_1 - g(\phi^{-1}(0))$, or constant when ϕ is. If K is not deterministic, constant kernels included, then for that fixed K some chamber contains margins on both composite sides.*

Proof. For a function ϕ, $h_1 = g(\phi^{-1}(0))$, so $f_1 - h_1$ is a wall function of (f, g), or f_1 or $f_1 - 1$ when $\phi^{-1}(0)$ is empty or everything. If some K_{j0} lies strictly between 0 and 1, the hyperplane $f_1 = \sum_j g_j K_{j0}$ is not a wall, and it meets the open parameter region, so it passes through the interior

of some chamber. For example, with $K = \left(\begin{smallmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \\ 0.6 & 0.4 \end{smallmatrix}\right)$, the margins $f = (0.5, 0.5)$, $g = (0.3, 0.3, 0.4)$ and $f = (0.515, 0.485)$, $g = (0.069, 0.463, 0.468)$ lie in one chamber, with $h_1 = 0.57$ above f_1 in the first and $h_1 = 0.436$ below it in the second. □

The interaction can be measured. There are 18 chambers. With f_1 uniform, g uniform on the simplex and the three kernel rows uniform, the composite side keeps 0.59 bits of uncertainty given the chamber; for the fixed kernel above, 0.28 bits; for a uniformly chosen deterministic kernel, 0 bits. The open shards, with the margins in a chamber and off the composite wall, number 36 for stochastic kernels and 36 for constant ones, since each chamber carries both composite sides, and 144 for the eight deterministic kernels, one side per chamber and kernel. So a joint transition, one in which the composite side changes inside a chamber, happens only for non-deterministic kernels, and it is exactly the information that the separate axes do not carry. This is the shard-level form of the integrality theorem of [3, Thm. 6.1]: deterministic composition is visible to the chamber structure, stochastic composition is not.

The same holds for 3×3 margins composed with a kernel $K\colon \{0,1,2\} \to \mathsf{D}(\{0,1,2\})$, where the composite is a 3×3 coupling with margins f and $K^{\mathsf{T}}g$ and 18 walls. There are 342 chambers, found by sampling and confirmed by closing the set under single wall crossings with linear programming. After any of the 27 deterministic kernels the composite chamber is a function of the original one. Under the sampling law the chamber carries 7.94 bits; given it, the composite chamber keeps 5.33 bits of uncertainty for a uniformly random kernel, 2.44 bits for the fixed kernel with diagonal 0.8 and off-diagonal 0.1, and 0 bits for a deterministic kernel. The open shards number 36,432 for stochastic kernels and as many for constant ones, the pairs of an original and a composite chamber that share a row margin, and $27 \times 342 = 9{,}234$ for deterministic kernels. For the 21 deterministic kernels that are not onto, the composite margin has zeros, so the composite coupling sits in the breaking regime although each of these shards is still one chamber.

6.5 Whiskering: the same picture one level up

The updates so far act on laws and margins. Whiskering acts on couplings: a kernel h on the second coordinate sends a coupling to the *product action* $\pi_h(\alpha) = h^{\mathsf{T}}\alpha h$ or to the *copy whiskering* $w_h(\alpha)$, which uses h once on equal pairs [1, §4], and the defect $F_2(\beta, \alpha) = w_h(\beta) \cdot w_h(\alpha) - w_h(\beta \cdot \alpha)$ measures how far w_h is from preserving gluing. Its absence from the dynamics so far is a matter of level: a band, a loop of updates or a shard is a statement about laws, and the defect is a statement about composing couplings. Read as updates on couplings, the two actions fall under the regimes of this paper.

Proposition 6.6 (Whiskering as an update)**.** *(i) The product action is the kernel update by $h \otimes h$ on the joint law, so proposition 6.4 applies: it heals when h has positive entries. The copy whiskering is strictly unital: it sends the diagonal coupling of f to the diagonal coupling of $h^{\mathsf{T}}f$, which stays at a vertex of its polytope, on the boundary, whatever h is.*

(ii) For $f = (\frac{1}{2}, \frac{1}{2})$, the couplings $\alpha_s = \left(\begin{smallmatrix} s & \frac{1}{2}-s \\ \frac{1}{2}-s & s \end{smallmatrix}\right)$ and β_r of the band $\mathrm{Cpl}(f, f)$, and $h_\varepsilon = \left(\begin{smallmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{smallmatrix}\right)$,

$$F_2(\beta_r, \alpha_s) = t(\varepsilon)\,(1-2s)(1-2r)\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad t(\varepsilon) = 2\varepsilon(1-\varepsilon)\big(\varepsilon^2 + (1-\varepsilon)^2\big).$$

The defect vanishes when either coupling is the unit, the diagonal $s = \frac{1}{2}$, and is largest at the opposite vertex, the swap. Along the kernel axis it vanishes exactly for the identity and the

swap, $\varepsilon \in \{0,1\}$, the coefficient t equals $\frac{369}{2500}$ at $\varepsilon = \frac{1}{10}$, and t is largest, $\frac{1}{4}$, at the constant kernel $\varepsilon = \frac{1}{2}$, as in [5, Prop. 4.3].

(iii) For a constant kernel with rows $q > 0$, $F_2(\beta, \alpha) = \big(\operatorname{tr}\alpha \operatorname{tr}\beta - \operatorname{tr}(\beta \cdot \alpha)\big)\big(\operatorname{diag}(q) - qq^{\mathsf{T}}\big)$, which is nonzero in general.

Proof. (i) $\pi_h(\alpha)$ is the law of $(h(y), h(y'))$ under α with independent copies. For w_h, each equal pair (y, y) is sent along one copy of h, so the diagonal goes to $\sum_y f_y \operatorname{diag}(h_y) = \operatorname{diag}(h^{\mathsf{T}} f)$. (ii) A direct computation with $w_h(\alpha) = \pi_h(\alpha) + \sum_y \alpha_{yy}(\operatorname{diag} h_y - h_y h_y^{\mathsf{T}})$ [1, Prop. 4.1], carried out symbolically in the checks. (iii) For constant h, $\pi_h(\alpha) = P$, the product coupling qq^{T} times the mass, and the copy correction is $\operatorname{tr}(\alpha)E$ with $E = \operatorname{diag}(q) - qq^{\mathsf{T}}$; since $P \cdot E = 0$ and $E \cdot E = E$ in the gluing algebra of q [1, Thm. 2.2], $w_h(\beta) \cdot w_h(\alpha) = P + \operatorname{tr}\alpha \operatorname{tr}\beta\, E$, while $w_h(\beta \cdot \alpha) = P + \operatorname{tr}(\beta \cdot \alpha)E$. □

So whiskering obeys the regime picture one level up, with one difference worth recording. For equivariant correspondences [3, Thm. 6.1], constant kernels behave like deterministic ones; for the walls of composites (proposition 6.5) they behave like stochastic ones; and for whiskering they are the worst case. Unitality, the reason w_h keeps the diagonal on the boundary, is also the reason it cannot be functorial for stochastic h [4, Thm. 10.2], and the defect is largest where the band of couplings is farthest from the unit.

7 Stability along paths

Two one-parameter families show what bends, what tears and what breaks, structure by structure.

Example 7.1 (A 2×2 path through two walls)**.** Fix $g = (0.3, 0.7)$ and let $f = (f_1, 1 - f_1)$ run over $(0, 1)$. The walls are $f_1 = 0.3$ and $f_1 = 0.7$, and the boundary points are 0 and 1.

structure	along the path	regime
gluing algebra $A_f \cong \mathbb{R} \times M_1(\mathbb{R})$	constant	bends throughout
product coupling $\alpha_{11} = 0.3 f_1$	linear	bends throughout
moment segment, length $\min(f_1, 0.3) - \max(0, f_1 - 0.7)$	kinks at 0.3 and 0.7, length $\to 0$ at the ends	tears, then breaks
nonzero torus brane, at the Fréchet midpoint	$f_1/2$, then 0.15, then $(f_1 - 0.4)/2$	tears at both walls
vertex entropies	$d \log(1/d)$ cusps at the walls	tear

On the walls themselves the margins are degenerate and the toric statements of [3] apply on either side. The product coupling and the nonzero brane are different points except at $f_1 = 0.5$ in the middle chamber, and the slope of the brane's location jumps by $\mp\frac{1}{2}$ at the two walls. As $f_1 \to 0$ the toric sphere $X_{f,g}$ shrinks to a point: the geometry breaks with the margin.

Example 7.2 (The three-bit path)**.** For the pairwise disagreement $1 - \varepsilon$ of [3, Ex. 3.3], the shards along $\varepsilon \in [0, 1]$ are: $[0, \frac{1}{3})$, incompatible, with contextual fraction $1 - 3\varepsilon$; $\{\frac{1}{3}\}$, compatible only on the boundary; $(\frac{1}{3}, 1)$, positive; and $\{1\}$, boundary again. On the positive shard the best minimum cell of a compatible law is $\min(\frac{3\varepsilon - 1}{4}, \frac{1 - \varepsilon}{4})$, attained by the maximum-entropy filler, so the filler's break margin falls linearly to 0 at both ends, while its Fisher form $\frac{8}{3\varepsilon - 1} + \frac{24}{1 - \varepsilon}$ blows up.

8 Second-order effects of breaking

A first-order break is a probability reaching 0. Its second-order effects are what happens, as a consequence, to structures built on top: global compatibility, the higher levels of the tower, sample sizes, algebras and branes.

Theorem 8.1 (Compatibility is lost only through breaking)**.** *Let $m(s)$ be a continuous path in the affine span of the achievable margins $Q = M(\mathsf{D}(X))$, compatible for $s < s^*$ and incompatible for $s > s^*$. Then $m(s^*)$ is compatible, every compatible law at s^* lies on the boundary of the simplex, and the largest minimum cell of a compatible law, $\delta(m) = \max_{\pi \in C_m} \min_x \pi_x$, tends to 0 as $s \uparrow s^*$.*

Proof. Q is a polytope, so it is closed and $m(s^*) \in Q$. If some compatible law at s^* had full support, then $m(s^*)$ would lie in $M(\mathsf{D}^\circ)$, which is the relative interior of Q because a linear map carries the relative interior of a polytope onto that of its image [25, Thm. 6.6]; the relative interior is open in the affine span, so $m(s) \in Q$ for s slightly above s^*, a contradiction. The function δ is concave on Q, since convex combinations of compatible laws are compatible for the combined margins, and upper semicontinuous, since C_m depends upper hemicontinuously on m; a concave function on a polytope is lower semicontinuous [19], so δ is continuous on Q, and $\delta(m(s^*)) = 0$. □

So a global section cannot disappear while the canonical filler is still strictly positive: the loss of the limit is always announced by a break of the filler. The next statement turns this into a margin.

Proposition 8.2 (Tolerance from the break margin)**.** *Let π be compatible with m and have minimum cell δ, and let $\kappa = \|M^+\|_{\infty\to\infty}$ for the pseudo-inverse M^+. Every change Δm of the margins within the column space of M that keeps the total mass of every table, with $\|\Delta m\|_\infty < \delta/\kappa$, leaves them compatible with a positive law. For pairwise tables of three bits $\kappa = \frac{3}{2}$, and on the path of example 7.2, for $\frac{1}{3} < \varepsilon \le \frac{1}{2}$, the bound $\frac{3\varepsilon-1}{6}$ equals the distance of the tables to the threshold.*

Proof. $\pi + M^+\Delta m$ has margins $m + \Delta m$, total mass 1 because the all-ones vector lies in the row space of M and every table keeps its mass, and all cells at least $\delta - \kappa\|\Delta m\|_\infty > 0$. On the path, $\delta = \frac{3\varepsilon-1}{4}$ and each table entry moves by $\frac{1}{2}$ per unit of ε. □

Remark 8.3 (The max–min coupling and the Floer side). For a single coupling polytope $\mathrm{Cpl}(f, g)$ the maximizer of δ is generally not unique, and refining it lexicographically, by maximizing the next smallest cell on the set of maximizers and so on, gives the leximin coupling. Taken over the facet cells, the cells that can vanish, this is the iterated max–min of Fukaya, Oh, Ohta and Ono [34, Thm. 1.5], whose torus fibre is nondisplaceable; the cells that cannot vanish are bounded below by $f_i + g_j - 1 > 0$. For 2×3 couplings, in every computed case it carries a nonzero torus brane, usually most of them, while the product coupling carries none whenever the smallest value $f_i g_j$ over the facet cells is attained at one cell [3, Thm. 8.2]. So in these cases the coupling that is lexicographically farthest from breaking, the quantity behind theorem 8.1 and proposition 8.2, is also the one the Floer theory singles out.

The second-order effects of a break are of five kinds.

(a) *Global compatibility* is lost only through a break (theorem 8.1), with a computable early warning (proposition 8.2).

(b) *Higher levels of the tower.* A zero at level one removes a model at level two for ever, and reliability at level three prevents it (section 9).

(c) *Sample sizes.* With joint samples, positivity at distance m from a break is confirmed, by seeing a rare cell, from $O(1/m)$ samples instead of $O(1/m^2)$, because the Fisher information diverges [3, §3]; with pairwise-only samples the rate stays $O(1/m^2)$.

(d) *Algebra and geometry.* Losing an outcome shrinks the gluing algebra from dimension $(n-1)^2+1$ to $(n-2)^2+1$; its Morita type, that of two points, changes only when the last circulation goes ($n = 2 \to 1$), where it becomes that of one point. The toric manifold loses dimension, as in example 7.1.

(e) *A break that does not cascade.* When a middle probability g_y of a chain tends to 0, the conditional row K_y of the second coupling becomes undefined, yet the glued coupling stays continuous: changing K_y moves it by at most g_y in total variation. The break damages the kernel, not the composite.

9 The tower under evidence

The tower's calibrated view of an outcome is a level-two law $\Pi = \sum_k w_k \delta_{\rho_k}$ over candidate models, its band $\mathrm{conv}\{\rho_k\}$ and its barycenter, the predictive law [7]. Evidence x updates the weights, $w_k \propto w_k \rho_k(x)$, which is conditioning at level two, and proposition 4.7 splits the entropy of the predictive law at every step into the expected entropy of the models and the epistemic term $I(K;X)$.

Example 9.1 (A break at level one removes a model for ever)**.** Take two models of a three-valued outcome (entropies in nats), $\rho_1 = (0.5, 0.5, 0)$ and $\rho_2 = (0.2, 0.3, 0.5)$, equal prior weights, and the observations $0, 1, 0, 1, 2, 0, 1, 0$. The weight of model 1 and the epistemic term evolve as follows, without and with linear discounting of each model towards the uniform law with reliability $\alpha = 0.9$:

observation	0	1	0	1	2	0	1	0
w_1, plain	0.714	0.806	0.912	0.946	0	0	0	0
$I(K;X)$, plain	0.214	0.186	0.120	0.088	0	0	0	0
w_1, discounted	0.694	0.783	0.891	0.929	0.473	0.671	0.764	0.880
$I(K;X)$, discounted	0.146	0.125	0.080	0.057	0.152	0.150	0.130	0.085

Without discounting, the first-order break in ρ_1 (the zero at outcome 2) becomes a second-order break of the tower: one observation removes the model that held 95% of the weight, the band collapses to a point, the epistemic term drops to 0, and no later evidence can bring the model back, however strongly it favours it. This is Cromwell's rule seen as breaking. With discounting, the surprise halves the weight and raises the epistemic term from 0.057 to 0.152, so the calibrated view registers the conflict as uncertainty, and the weight recovers to 0.88 as the later observations favour model 1.

Proposition 9.2 (Reliability bounds how fast the calibrated view can morph)**.** *If every model is linearly discounted towards the uniform law on n outcomes with reliability $\alpha < 1$, then each observation changes the odds between two models by a factor at most $1 + \alpha n/(1-\alpha) \le n/(1-\alpha)$. So no weight reaches 0 after finitely many observations, and the band of the calibrated view, the hull of the discounted models, stays at distance at least $(1-\alpha)/n$ from every face of the simplex: it cannot break. Weights may still tend to 0 as evidence accumulates.*

Proof. Discounted probabilities lie between $(1-\alpha)/n$ and $\alpha + (1-\alpha)/n$, and the odds change by the ratio of two of them. □

Read through the tower, the three regimes of the paper become three behaviours of the calibrated view: it bends while evidence moves the weights, it tears when the predictive law crosses a decision wall, and it breaks when a level-one zero removes a model. Reliability at level three converts breaking into bending, at the price of a bounded speed.

Remark 9.3 (Floors as policies). Discounting imposes a floor $(1-\alpha)/n$ on every outcome. For a floor τ on cells of a two-way table, the floored coupling polytope is a translate of the coupling polytope of the reduced margins $f - \tau\mathbf{1}$, $g - \mathbf{1}^\top\tau$ [4, Prop. 6.9]. Its walls are those of the reduced margins, so a policy alone can tear or break a system, and its headroom, the largest common margin above the floors, is the break margin of definition 3.1 computed on the reduced margins. Proposition 9.2 bounds the law, not a model fitted to finitely many samples: in recursive training with known truth a fitted model still assigns zero to rare categories at every share of fresh data [35, Section 13]. Along a chain of transports a floor is never lost, only weakened where a set is summarised by its floors [7, Thm. 10.26].

10 Evidence I: two pilots on public data

10.1 Subgroups as contexts: NHANES

Data. NHANES 2009–10 and 2011–12, as distributed in the R package `NHANES` [22]: 10,730 adults aged 20 or more with blood pressure, body mass index and diabetes status, with examination weights and a stratified jackknife over primary sampling units for standard errors. Four attributes define 16 subgroups: age 55 or more, diabetes, overweight (BMI ≥ 25) and male sex. The outcome is high systolic pressure (≥ 140 mmHg), with weighted prevalence 12.8%.

The information ladder. Suppose the sizes of all 16 subgroups and the overall prevalence are known, and the outcome is known otherwise only through tables of the outcome with k of the attributes. The risk of a subgroup then lies in a band, computed by linear-fractional programming over all joints with these tables. For the subgroup "55+, diabetic, overweight, male" (462 adults, risk 0.308):

information	band for the subgroup	mean width, all 16 subgroups
sizes and overall prevalence	[0, 1]	0.95
outcome by one attribute at a time	[0, 0.881]	0.73
outcome by pairs of attributes	[0.170, 0.430]	0.33
outcome by triples of attributes	[0.302, 0.311]	0.04

This is the precision ladder of [2, Thm. 3.5] on real data. Every band contains the subgroup's risk, but that holds by construction, since the bands are computed from the same joint table, and is only a sanity check. The test is out of sample: build the bands from the 2009–10 tables and ask whether the 2011–12 risks fall in them. With one or two attributes at a time 16 and 15 of the 16 new risks fall inside. With triples none do: those bands are narrower than the sampling variation between cycles. When each new risk is given its own 95% sampling interval, 12 of 16 intervals meet the triple-level bands. A structural band alone is overconfident once it is narrower than the sampling noise, which is the point of the next paragraph.

Two kinds of imprecision. With a risk threshold of 0.30 as the wall, the structural band from the triples lies entirely above the wall, while the sampling band, $[0.25, 0.37]$, straddles it. (Sampling

bands are Wilson intervals with the design-based effective sample size $r(1-r)/\mathrm{se}^2$.) A tolerance measure has to carry both, the structural band of the information available and the sampling band of the estimate.

Revision. Between the two survey cycles the subgroup's risk moved from 0.25 to 0.36, outside its earlier 95% band. Of the 16 subgroups, 11 landed inside their earlier bands. With the actual standard errors of both cycles, about 11.3 of the 15 subgroups with positive standard errors would be expected inside if nothing had changed, so 11 is consistent with no change. Four subgroups changed status relative to the wall, between decisive and straddling, and none crossed it. One subgroup had 7 members and no cases in 2009–10; its survey standard error is exactly 0, and the band collapses to a point. This is the breaking regime in practice: at the boundary of the simplex the design-based band is undefined, and we fall back to a count-based interval, $[0, 0.35]$.

Contexts and incompatibility. Pairwise tables taken alternately from the two cycles are slightly incompatible, with least L^1 adjustment 0.084, but less than tables taken alternately from two halves of the 2009–10 cycle (0.179 in the median of 20 random splits of its sampling units). Halves have half the sample size, which inflates their noise by about $\sqrt{2}$, and $0.179/\sqrt{2} \approx 0.13$ still exceeds 0.084. So the incompatibility is no larger than sampling noise, and for these tables the two cycles behave as one population (section 11). There is no full Simpson reversal among the attributes, but the effect of being overweight changes sign with age: +4.5 points under 55 (standard error 0.7) and −3.3 points at 55 or more (1.8), clear in the first stratum and borderline in the second.

Barcodes of the ladder. For each subgroup and each risk threshold, the band at information level k either meets both sides of the threshold or lies on one side; the *decision level* is the first k at which it lies on one side, and the bar of the losing side ends there. At the threshold 0.30, the 16 subgroups are decided at levels 1, 2 and 3 by 3, 3 and 9 subgroups, and one is never decided; at 0.20 by 1, 3 and 10, with two never decided; at 0.10 at levels 2 and 3 by 9 and 5, with two never decided. In sample the decided side is always the true one, by construction. Out of sample, with bands built from 2009–10 and judged against the 2011–12 estimates, the decisions made at levels 1 and 2 are right 18 times out of 18, and those made only at level 3 are right 23 times out of 30. A decision that waits for the top of the ladder rests on a band narrower than the sampling noise, which is the overconfidence seen above.

10.2 Revised forecasts and revised data: the COVID-19 Forecast Hub

Data. The ensemble forecasts of the US COVID-19 Forecast Hub [12]: weekly incident deaths for 56 states and territories, forecast dates June 7, 2021 to March 21, 2022 (targets ending by March 26, so that every target has six weeks of later data revisions), horizons one to four weeks, 9,072 forecasts given by 23 quantiles. The truth file of the hub's repository was read at every forecast date, so we know what was known when; the final truth is the last state-level version (May 9, 2022). The band is the central 80% interval, and each state's wall is its median weekly deaths over the period.

Calibration and movement. The bands cover the final value 78–80% of the time at every horizon. Revising a forecast for a fixed week, as the horizon shrinks from four weeks to one, widens the band 26% of the time, and 64% of the time the new band is not contained in the old one: information moves bands as well as shrinking them.

Tear time. A band is decisive if it lies on one side of the wall, and it tears when its next revision is not on that side. The analogue of the tear time used here is one-dimensional: τ is the margin from the band to the wall, in deaths, divided by the latest movement of the median, in deaths per revision. It is defined for the 1,954 decisive bands that have a later revision, an earlier one, and a median that moved; the four-week forecasts (936) have no earlier revision and 253 medians did not move.

τ (weeks)	< 0.5	0.5–1	1–2	2–4	≥ 4
share that tear at the next revision	0.33	0.20	0.11	0.10	0.03

The ranking of decisive bands by τ separates tearing from not tearing with area under the curve 0.78. The relative margin alone does slightly better, 0.80. So the speed term of τ adds nothing in this pilot, perhaps because a single movement is a noisy estimate of speed.

Crossing by straddling. Of all 3,143 decisive bands with a later revision, only 3 jumped from one side of the wall to the other. Bands cross walls by passing through a straddling state, the discrete counterpart of theorem 3.2.

Unreliable data. Of 2,408 state-weeks, 3.2% were later revised by more than 25%. One-week-ahead forecasts made when the latest week was such a week covered the final value 43% of the time (95% interval 33–55%), against 79% otherwise (74 and 2,278 forecasts). The bands did not account for the possibility that their inputs would be revised. This is the strongest empirical reason for a reliability level above the band.

10.3 What the pilots support

They support the band picture in four ways: bands narrow with information and, combined with sampling bands, predict new data; decisive bands rarely jump across walls; small margins predict tearing; and unreliable inputs break calibrated bands. They do not yet support the tear time as more than a margin, and they do not test holonomy or dilation in the strict sense; the 26% of forecast bands that widen are not dilations, because successive forecasts condition on different information.

11 Evidence II: a protocol for clinical data

Clinical records, for example MIMIC-IV [21], offer transitions between patient states, measurements made on different subsets of patients, and timestamps for when each value was taken and when it was entered. The protocol below separates the statements that are theorems, which can only fail through errors of code or estimation, from those that could be false.

1. *Conditional reconstruction.* Estimate transition kernels between clinical states and validate them on held-out patients and on a later period. This is a prerequisite, not evidence.

2. *Coupling geometry.* The dimension $(n-1)(m-1)$, the circulation space and the vertices of the coupling polytope follow from the margins. The falsifiable part is where the observed joint sits: its tear margin, whether it lies on a face (structural zeros), and whether patients whose margins lie near a wall are those whose classification changes between periods.

3. *The canonical filler.* By proposition 4.8 the filler always maximizes entropy in its fibre. What can fail is the claim that the gap $I(X; Z \mid Y)$ is small when Y is the clinically meaningful intermediate state; it is tested by $2N\, I(X; Z \mid Y)$, asymptotically χ^2 with $(|X| - 1)(|Z| - 1)|Y|$ degrees of freedom.

4. *Higher-order compatibility.* If all tables come from the same patients, a common joint exists: the empirical one. So incompatibility can only come from tables computed on different patients, which in clinical data mostly means available-case analysis of tests ordered for sicker patients. The observed incompatibility is split into sampling (a simulated or bootstrap null), selection (the part removed by inverse-probability weighting for being measured) and a residual that points to differences between populations. The measure is fixed in advance: the contextual fraction of Abramsky, Barbosa and Mansfield [9], one minus the largest weight of a sub-normalized common joint below the tables, with the least L^1 adjustment as a second measure. A successful filling does not count against the theory.

5. *The posterior tower.* That kernels preserve coherence and can only lower the secondary obstruction is a theorem [7, Prop. 9.14], [4, Prop. 11.5]. The falsifiable claim is that the estimated obstruction, tear margin or tear time predicts deterioration or intervention beyond a standard severity score.

6. *Information arrival.* The two timestamps of each measurement reconstruct what was known at each moment, which is the test of section 10.2 at the level of a patient.

The thresholds that define walls, the outcomes and the incompatibility measure are to be fixed before the data are examined. The credentialed data may not be sent to online services unless their handling meets the data use agreement [23], so the analysis is meant to run where the data are held.

Synthetic validation. Each test was run on synthetic data with a planted answer (checks S1–S3).

- *Test 3.* With $N = 2000$ and $X \perp Z \mid Y$ the G-test rejects at level 0.05 in 4.8% of 400 replicates; with a planted effect of 8 points it rejects in 88% of 200.
- *Test 4.* A lab measured with probability 0.9, 0.55 or 0.2 according to a covariate makes the available-case tables incompatible: least L^1 adjustment 0.439 and contextual fraction 0.220, against a sampling null with median 0.021 and 95% quantile 0.040 when the lab is missing completely at random. Weighting by the inverse of the estimated probability of measurement brings the contextual fraction of the same sample to 0.021, about the 95% quantile (0.020) of the contextual fraction under the sampling null, and over repeated samples the weighted inconsistency has median 0.015 and 95% quantile 0.042, the same order as the sampling null. The decomposition recovers the planted selection.
- *Walls.* For margins with tear margin 0.080, the chamber is misidentified from N samples in 24%, 0% and 0% of 300 replicates at $Nm_T^2 = 0.5$, 4 and 16. The bound of [4, Prop. 12.2], about $Nm_T^2 \geq 11$ here for 95%, is conservative.

12 Open problems

1. *Holonomy.* Example 5.4 gives a residue from discounting that depends on where it is applied, not on the loop. Find a class of updates for which the residue depends only on the homotopy class of the loop, or show that none exists among updates of finite laws.

2. *A better tear time.* Estimate the speed from several revisions, and add the reliability of the inputs, to see whether τ then beats the margin in the forecast data.

3. *Persistence over band width.* The barcodes of section 10.1 use one threshold at a time. Compute the persistence module of the full chamber complex as a band widens, with several walls at once, and the barcodes for the forecast pilot.

4. *Bands of bands.* Carry the three regimes to levels two and three of the tower, where a band is a set of laws on laws and reliability is itself uncertain.

5. *Fisher–Rao margins.* Replace total variation by the Fisher–Rao distance, in which the boundary is at finite distance and the break margin and the tear margin live in one metric.

6. *Degeneration of the toric family.* Along a path of margins, describe the limit of the toric manifold, its symplectic potential and its nonzero branes at a wall and at the boundary, extending example 7.1; and decide whether the positive real Lagrangian of [3] extends to a compactification that keeps boundary-supported couplings.

7. *The interaction nerve.* Proposition 6.5 counts the open shards for 2×3 margins with a kernel axis and measures the information in joint transitions. Compute the full nerve, with the wall strata and the closures of the kernel strata, and its homotopy type.

8. *The five papers.* Add to each of the other papers a short subsection on its structures under bands.

13 Computational checks

All code is at `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/bands`. `check_bands.py` runs seventeen checks and `check_dynamics.py` eighteen more, all passing. The first:

- B1–B3: propositions 2.2 to 2.4, by linear programming;
- B4, B13: theorem 3.2 and proposition 3.4;
- B5–B8: propositions 4.2 to 4.5, 4.7 and 4.8;
- B9–B12: propositions 5.1 to 5.3 and 5.8;
- S1–S3: the synthetic validation of section 11;
- P1: the quoted pilot numbers against the saved pilot outputs, and the synthetic and example numbers against this run.

The second: D1 (proposition 6.1); D2 (proposition 6.4); D3 and D4 (theorem 8.1 and proposition 8.2, on random paths and perturbations); D5 and D6 (examples 7.1 and 7.2); D7 and D8 (items (e) and (d) of section 8); D9 (example 9.1 and proposition 9.2, including the table); N1 and N2 (proposition 6.5 and the counts after it, by the script `nerve/interaction.py`); N3 (the barcodes of section 10.1, by `nerve/barcodes.py`); N4 (the 3×3 counts, from the saved output of `nerve/interaction33.py`, which runs in about six minutes); W1 (proposition 6.6, symbolically); A1 (example 5.4); M1 (proposition 5.6, associativity of the Kleisli extension and the monad morphism, by support functions); H1 (proposition 6.2: random 2-horns, the 3-horn and 2-boundary counterexamples by linear programming, and 3-horns cut from random joint laws); G1 (remark 3.5: the GIT walls of the configuration $e_i + e_{n+j}$ for 2×2, 2×3, 3×3, exactly). Both run `extra/horns_git.py`. The pilots are reproduced by the scripts `nhanes_pilot.py`, `fhub_fetch.py` and `fhub_pilot.py` in the folder `pilot`.

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